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[22CSE120]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2023
ESSENTIALS OF DATA WAREHOUSING AND DATA MINING

Time: 3 Hours Maximum **Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	What are the different tasks of Data Mining?	L-1	CO5	**02**
	b.	What are the key features of Data Mining?	L-1	CO5	**02**
	c.	Explain Decision Support System or Business Intelligence?	L-2	CO4	**02**
	d.	What is a Dimension Table?	L-1	CO1	**02**
	e.	State the role of Metadata in a data warehouse?	L-1	CO1	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Explain univariate analysis on numerical and categorical features separately.	L-3	CO5	**05**
		OR			
	b.	Explain bivariate analysis on numerical and categorical features separately.	L-3	CO5	**05**
3.	**a.**	What is vertical partitioning? What are the scenarios where vertical partitioning can be used? Explain with examples.	L-2	CO4	**05**
		OR			
	b.	What is partitioning? State why partitioning is required with respect to data warehouses.	L-2	CO4	**05**
4.	**a.**	Describe OnLine Analytical Processing (OLAP) and its operations. State the characteristics of OLAP.	L-1	CO3	**05**
		OR			
	b.	Explain data loading and its challenges in a data warehouse.	L-2	CO2	**05**
5.	**a.**	Describe the challenges with extracting the data from an operational system onto the data warehouse.	L-2	CO2	**05**
		OR			
	b.	What is an enterprise warehouse? Explain with an architecture diagram. State how enterprise warehouse is different from data mart based data warehouse.	L-1	CO1	**05**
6.	**a.**	Explain the applications of Data Mining with examples?	L-1	CO5	**05**
		OR			
	b.	What is the difference between data cleaning and data scrubbing? Explain with examples.	L-2	CO2	**05**

7.	a.	Why is data staging so important before storing the data storage space?	L-1	CO1	05
		OR			
	b.	Explain steps involved in the ETL process in detail?	L-1	CO2	05
		PART-C	**6 X 10=60**		
8.	a.	State the architecture of snowflake schema using the case study of product sales with fact table: sales{ProductID, CustomerID, OrderID, EmployeeID, Total, Quantity, Discount}, dimension tables: Time dimension, Product dimension and ProductCategory dimension, Customer dimension and Address dimension, Employee dimension and Department dimension.	L-4	CO4	10
		OR			
	b.	What is fact constellation schema? State how is fact constellation schema different from snowflake schema in terms of their characteristics?	L-2	CO4	10
9.	a.	What is a data cube? Explain how a data cube can be used for data visualization.	L-3	CO3	10
		OR			
	b.	What is relational online analytical processing, multidimensional online analytical processing, and hybrid online analytical processing.	L-1	CO3	10
10.	a.	Discuss the architecture of virtual warehouses. State how two tier OLAP can work with virtual warehouse with example.	L-3	CO3	10
		OR			
	b.	State what and how dual level of granularity. Present a case study with its benefits and flaws.	L-2	CO1	10
11.	a.	What is a data mart? Explain the difference between an enterprise data warehouse and a data warehouse with data mart.	L-1	CO1	10
		OR			
	b.	What is the difference between a fact table and dimension table?	L-1	CO4	10
12.	a.	What are the various data transformation terms?	L-1	CO2	10
		OR			
	b.	What is data granularity? How granularity in the data warehouse works?	L-2	CO1	10
13.	a.	What are the characteristics and features of a snowflake schema?	L-1	CO4	10
		OR			
	b.	What are the characteristics and features of a star schema?	L-1	CO4	10